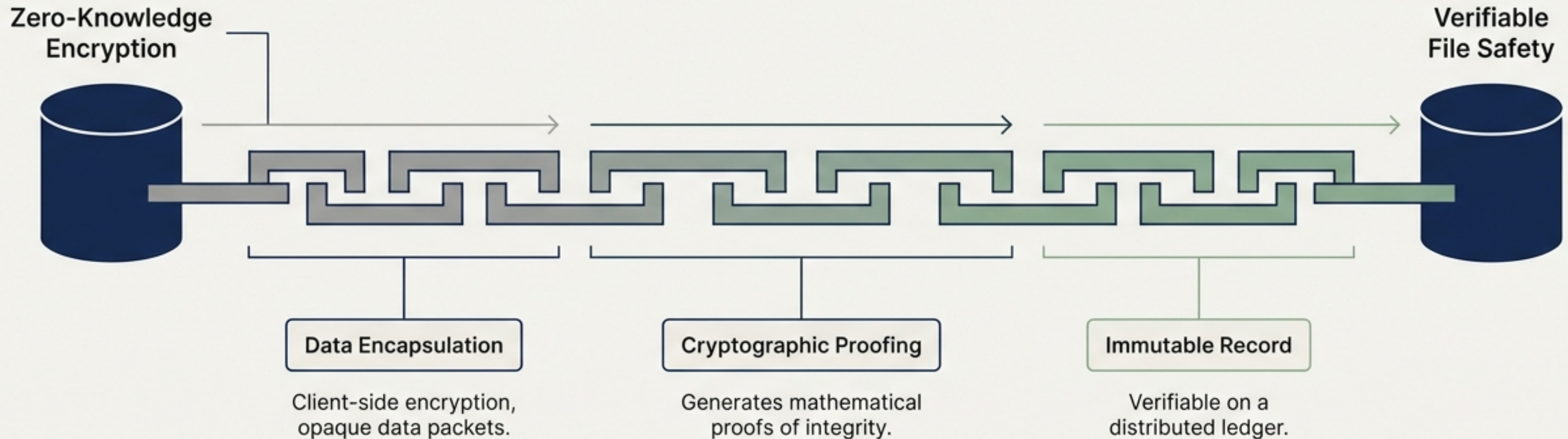


File Safety Orchestration Architecture

The bridge between zero-knowledge encryption and
cryptographically verifiable file safety.

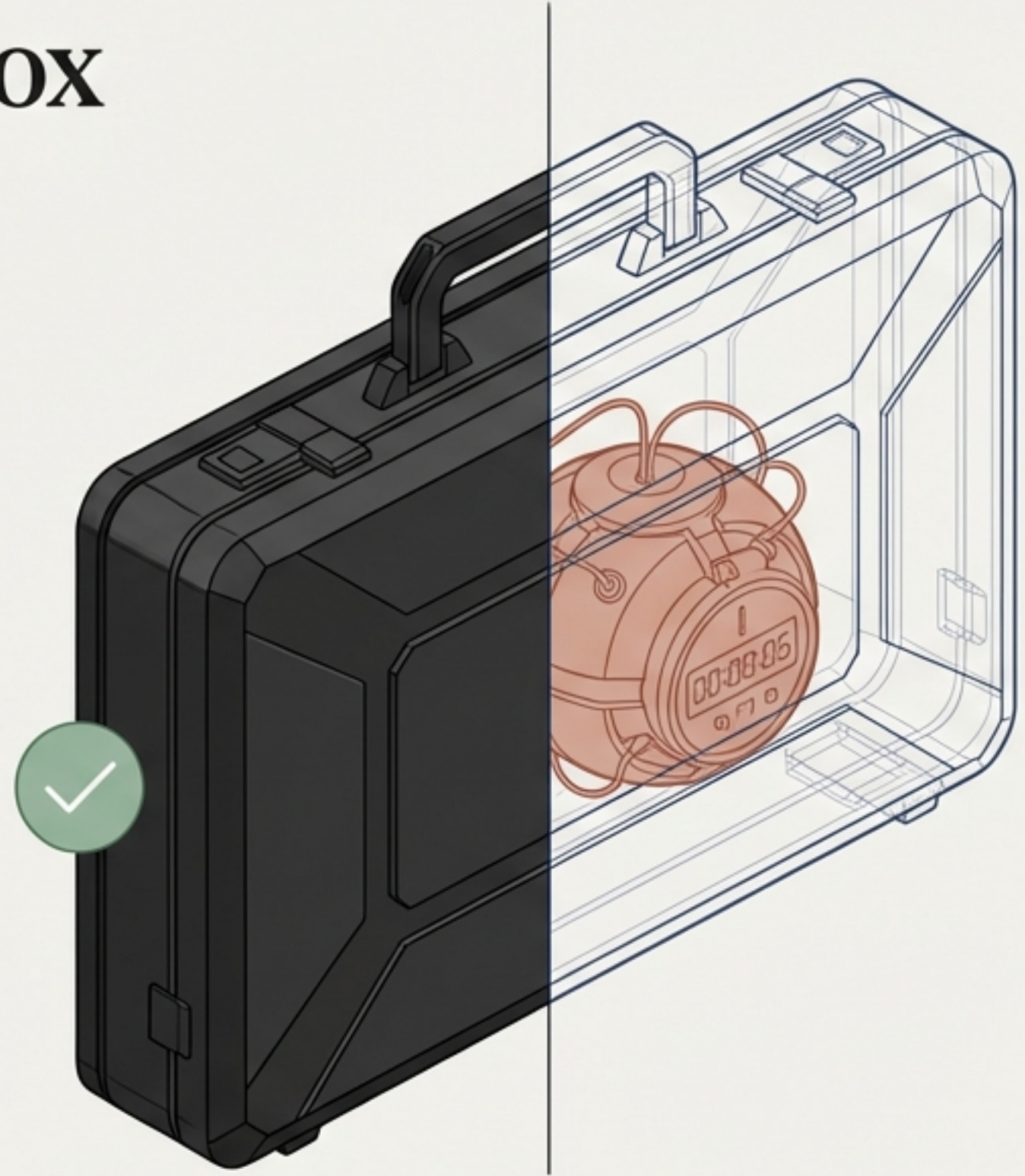


The Zero-Knowledge Paradox

Delivering a file securely does not make the file itself safe.

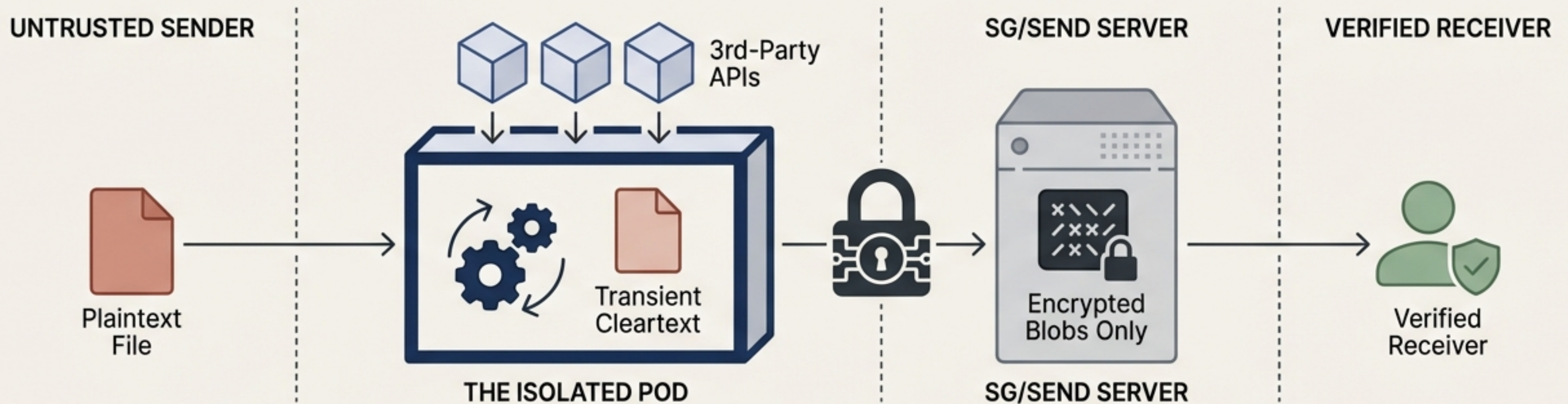
- SG/Send is a zero-knowledge encrypted platform. The server stores encrypted blobs it cannot read.
- A malicious sender can encrypt malware, and the server will deliver it faithfully.
- SG/Send is the secure courier. It does not inspect the package.

The Challenge: How do we scan what the server cannot see?



The PKI-Verified Scanning Chain

SG/Send cannot scan files, but trusted third-party services can. The results are cryptographically bound to the file via hash and encrypted exclusively for the receiver.



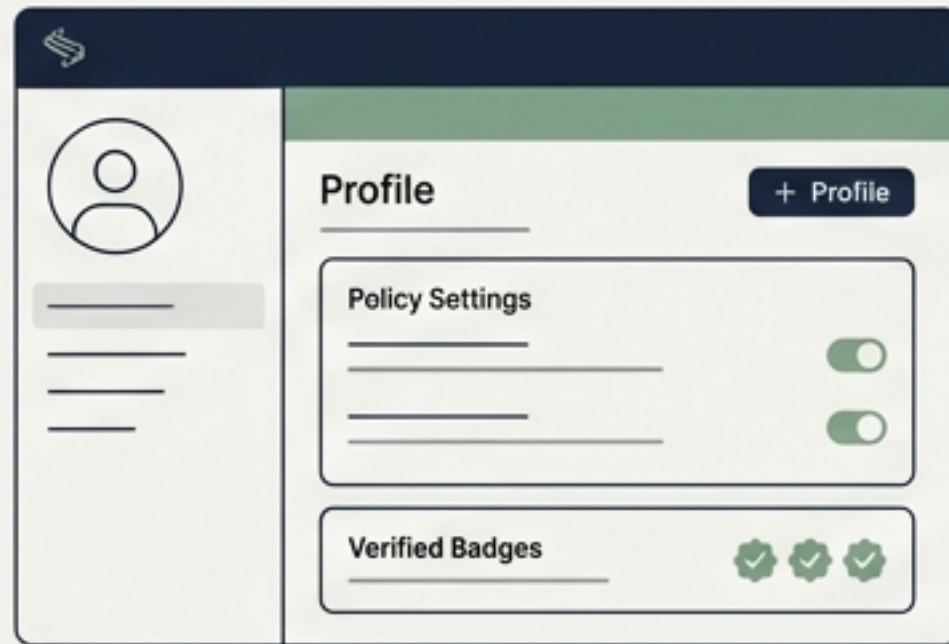
- **A safe file, delivered safely:** Scanning services rebuild or verify the file.
- **Isolated Orchestration:** A minimal, isolated Pod manages the scan, not the sender.
- **Cryptographic Proof:** The receiver verifies the exact byte-for-byte output of the scanning service before opening.

The Spectrum of File Safety

 CDR (Content Disarm & Reconstruction): Rebuilds known-good versions, stripping macros and polyglots (e.g., .	 Antivirus / Anti-malware: Signature-based detection of known threats (e.g., .	 File Reputation: Hash lookups against threat intelligence databases.	 DLP (Data Loss Prevention): Scans for PII, financial data, and classified markers.
 Sandbox / Detonation: Behavioral analysis in isolated environments.	 Metadata Stripping: Removes hidden forensics (GPS, author names, Exif).	 Format Validation: Detects polyglot files and type confusion attacks.	The Pod chains multiple services (e.g., CDR + AV + DLP) in a single pass, independently attesting each result.

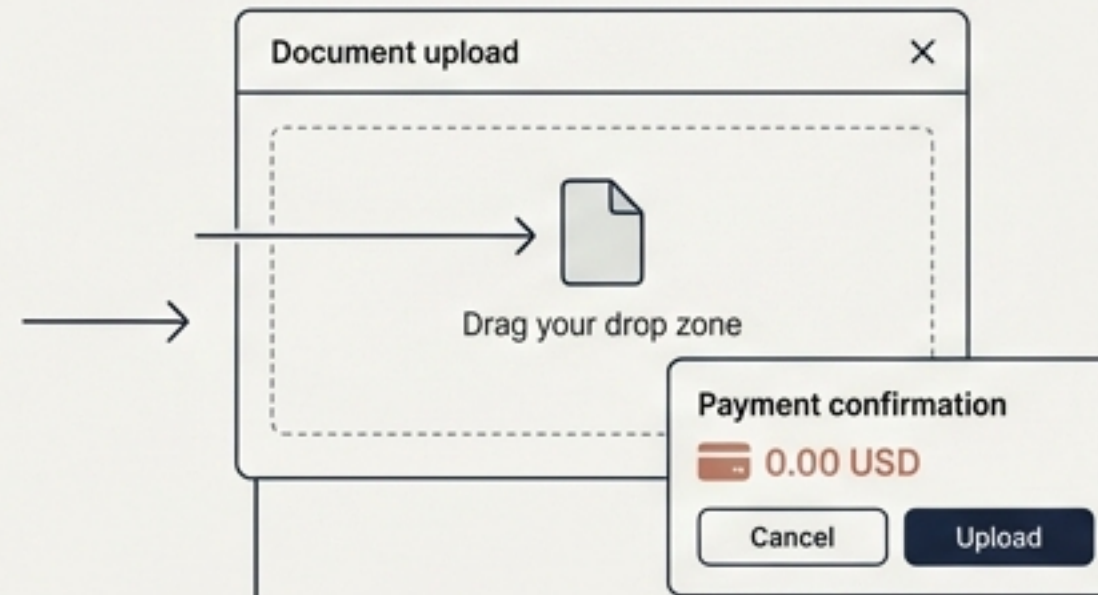
Inverting Procurement: A Pull-Based Economy

Procurement is no longer the bottleneck. Receivers mandate the security standard; senders absorb the micro-cost.



- **Receiver Defines Policy:**

Organizations publish delivery requirements in their SG/Send data room. (Requires CDR + AV)



- **Sender Pays Per Scan:**

Senders pay a micro-transaction to satisfy the receiver's inbound requirements.



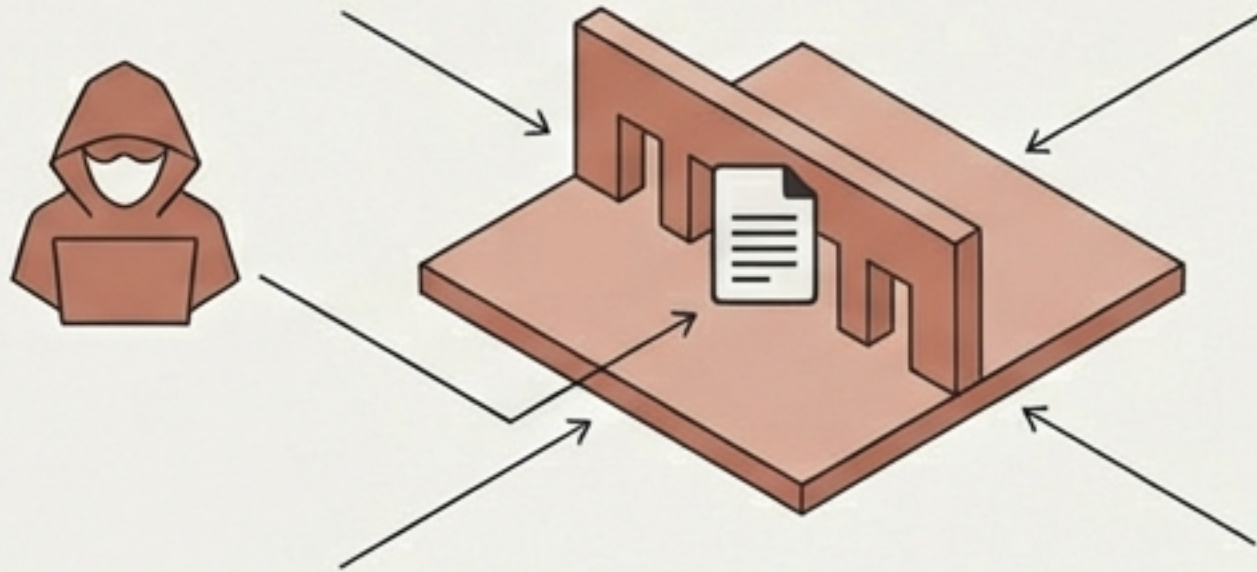
- **Automatic Partner Revenue:**

As the SG/Send network grows, scanning partners generate recurring API revenue automatically—without enterprise sales efforts.

The Cross-Domain Paradigm Shift

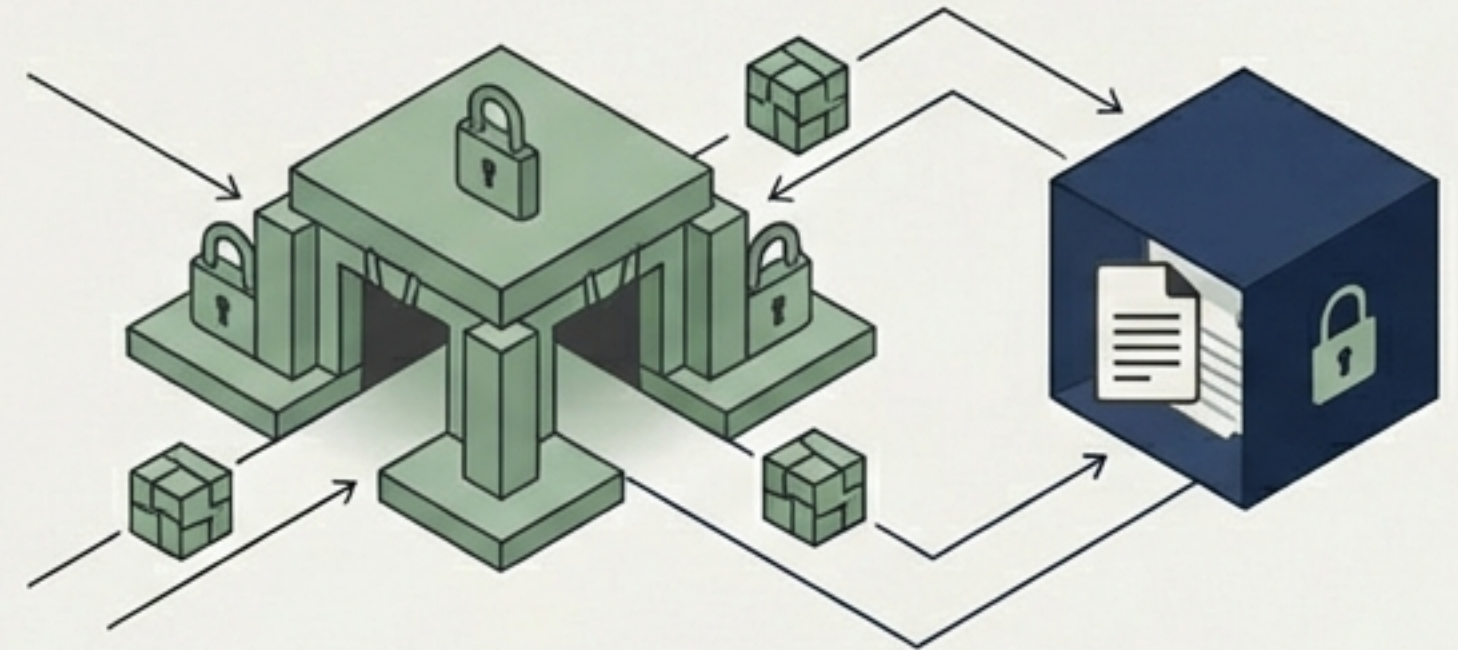
The network boundary never sees plaintext—ever. Not transiently, not in a buffer, not in a log.

Traditional



- Crossing point holds plaintext, making it a high-value attack target. Trust is established between domains.

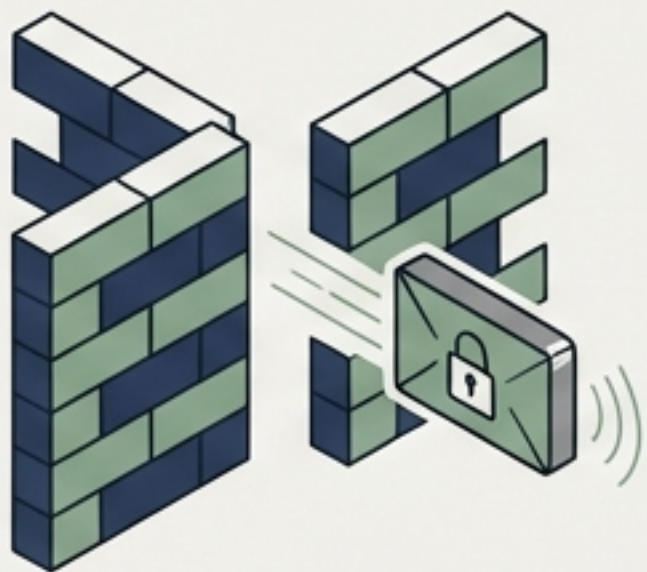
SG/Send



- Crossing point processes strictly encrypted blobs. The isolated Pod is in a separate network.
- Trust is established cryptographically with the Pod's public key.

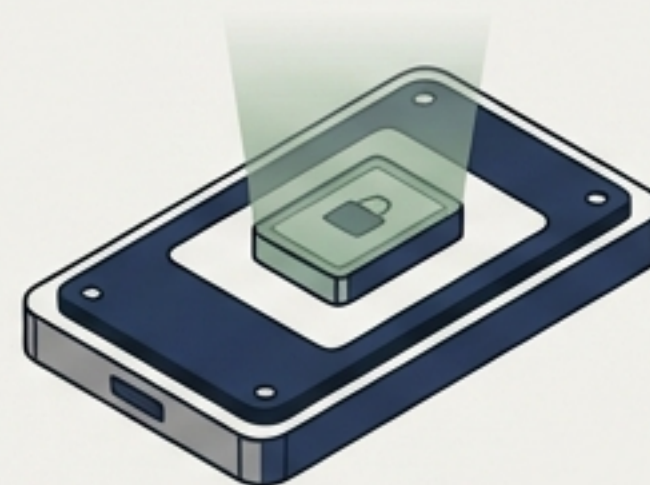
Impact: A breach of the crossing infrastructure exposes absolutely zero file content.

Engineered for Government & Defence



Cross-Domain CDR

Meets strict boundary mandates without the crossing infrastructure requiring access to plaintext.



Air-Gapped Operation

Embedded scanning engines support the offline .sgroom portable data room for disconnected supply chains.



Zero-Knowledge Audit

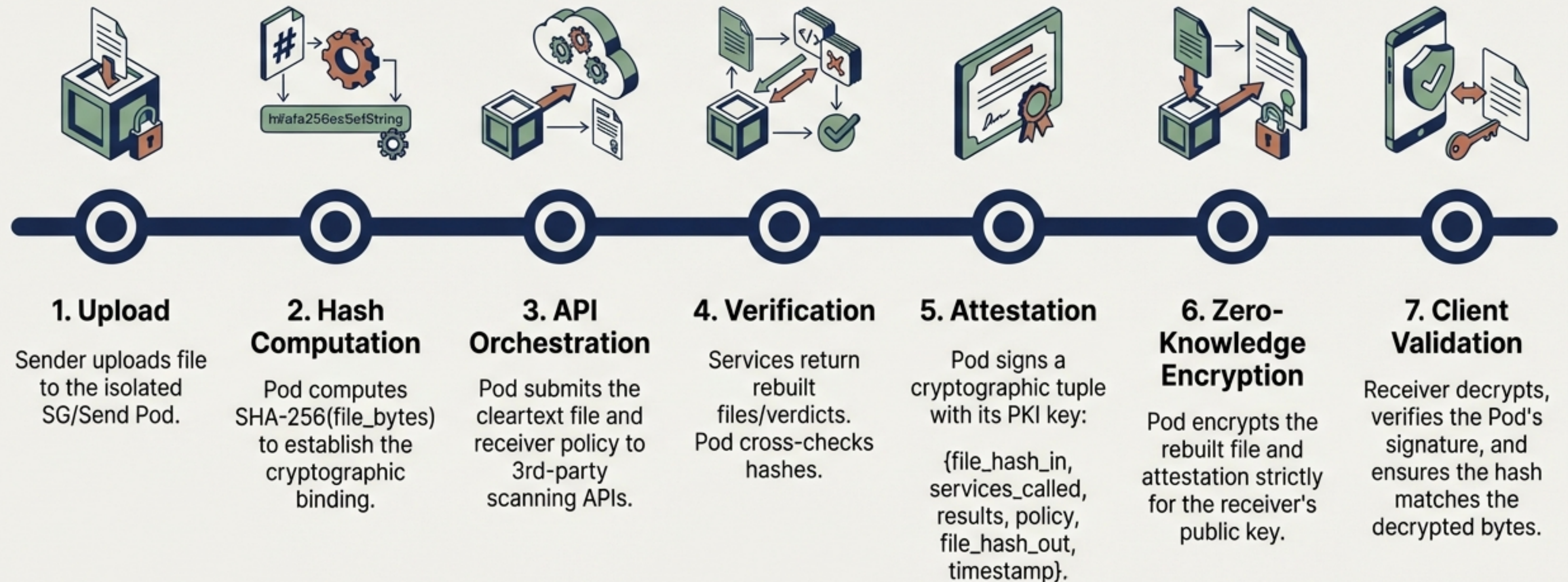
Generates PKI-signed audit trails of every transfer and scan result, without the server accumulating classified content.



Open-Source Verifiability

Built on an Apache 2.0 open-source codebase. No black boxes; fully auditable by national security technical teams.

Technical Architecture: The End-to-End Flow



Security by Design: The Untrusted Sender

The sender is explicitly untrusted. The Pod architecture removes the sender from all trust-bearing operations.



Threat: Swapping Files.



Mitigation: Sender never touches scanning APIs directly. The Pod hashes, scans, and re-encrypts in one isolated motion.



Threat: Forging Scan Results.



Mitigation: Attestations are signed with the Pod's private key (and eventually the scanning service's native key).



Threat: Skipping Scans.



Mitigation: Receiver clients strictly enforce valid attestations. No signature = file rejected before opening.

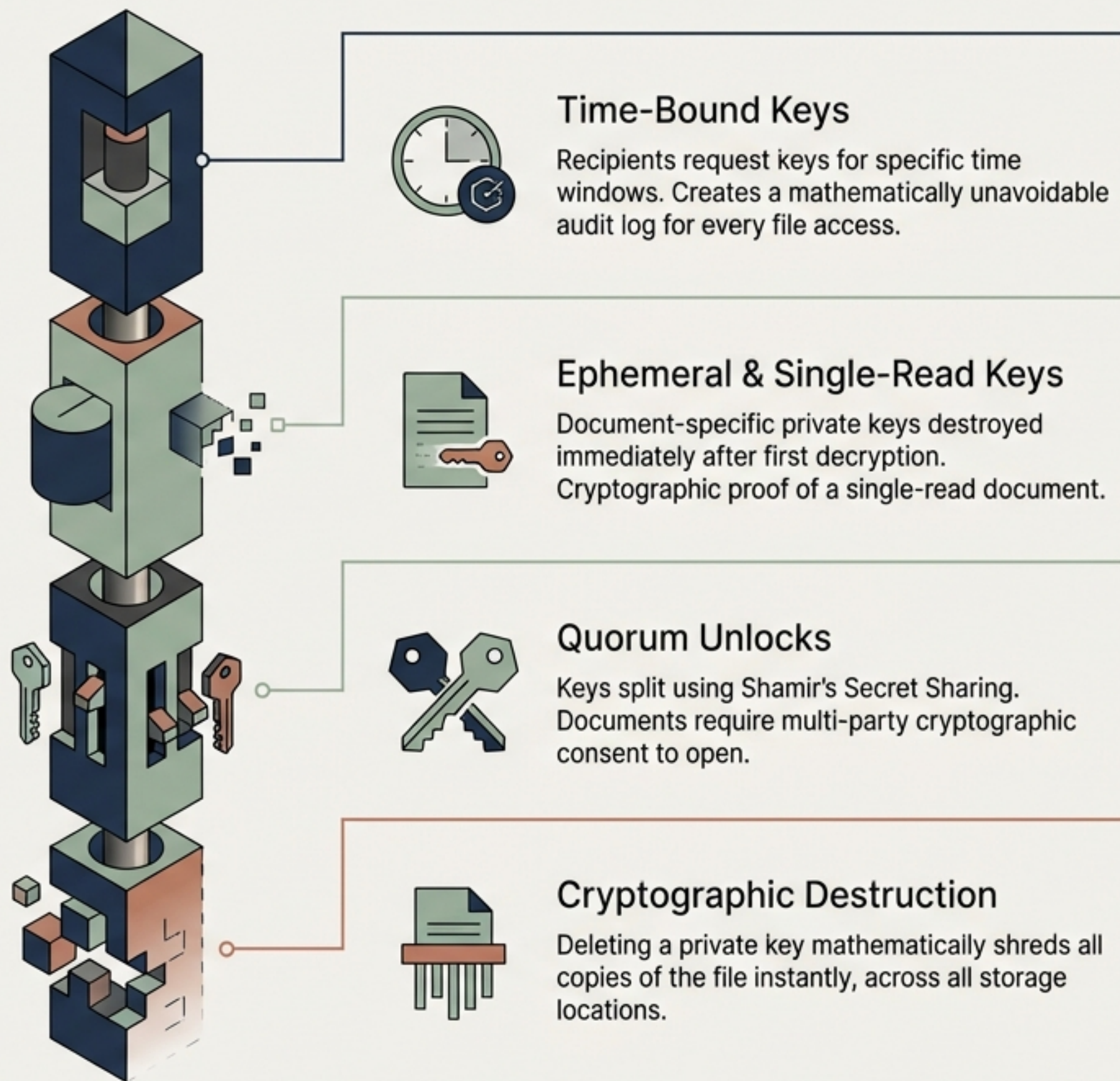


Threat: Replaying Old Scans.

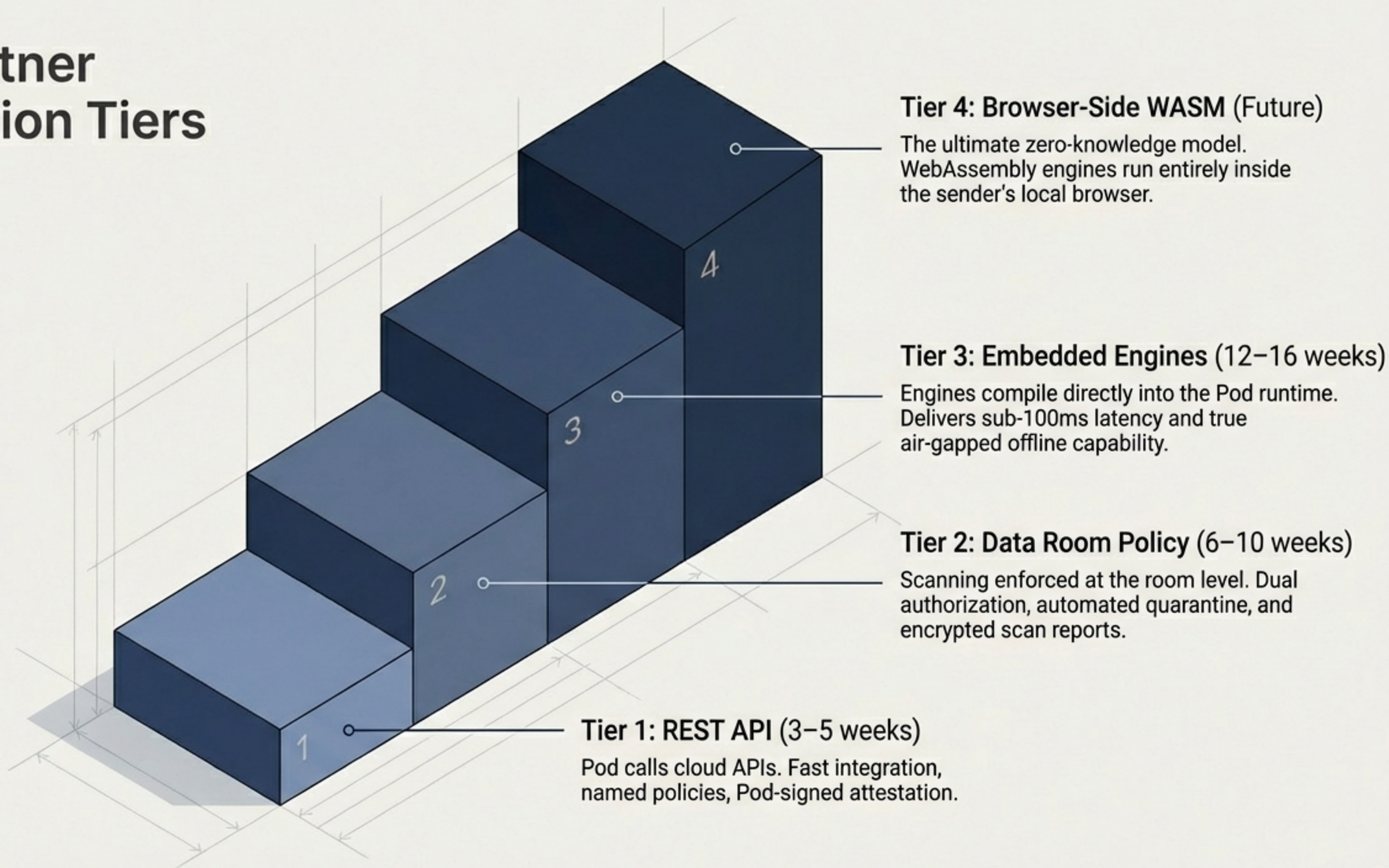


Mitigation: The cryptographic tuple includes exact input/output hashes and recency timestamps.

Advanced Cryptographic Control



The Partner Integration Tiers

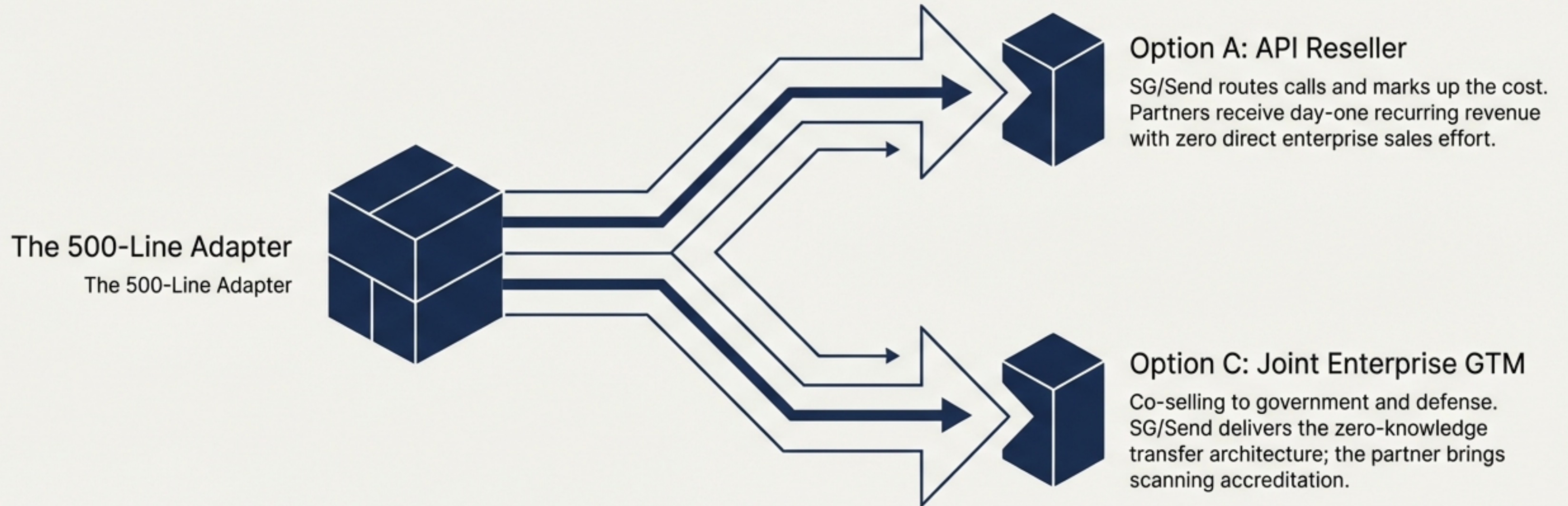


The Partner Ecosystem Menu

Basic (Included)	Standard	CDR	Enterprise	Government
Open-Source AV (ClamAV) + Format Validation. Establishes the scanning chain as default out-of-the-box infrastructure.	Commercial Multi-Engine AV + File Reputation. Low-cost per scan.	Deep Content Disarm & Reconstruction. Medium cost per scan.	CDR + Multi-AV + DLP + Sandbox. Designed for regulated finance and legal sectors.	Full stack + Embedded Air-Gapped Engines. Enterprise licensing for classified networks.

Partner Integration & Commercial GTM

Provide API access. We build the 200-500 line adapter. You access a new pull-based distribution channel.



Roadmap to Implementation

Phase 1 (Weeks 1-4)

ClamAV Baseline. Prove the end-to-end scanning chain without commercial dependencies.

Phase 3 (Weeks 8-14)

Data Room Policy Enforcement. Admin UI, quarantine workflows, room-level enforcement.

Phase 5 (Weeks 20-30)

Embedded Engines. Air-gapped offline support for defense networks.

Phase 2 (Weeks 4-8)

First Commercial CDR. REST API integration with sender-pays billing.

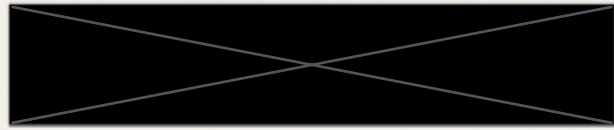
Phase 4 (Weeks 14-20)

Multi-Service Chaining. Orchestrating CDR + AV + DLP in a single attestation.

Phase 6 (Ongoing)

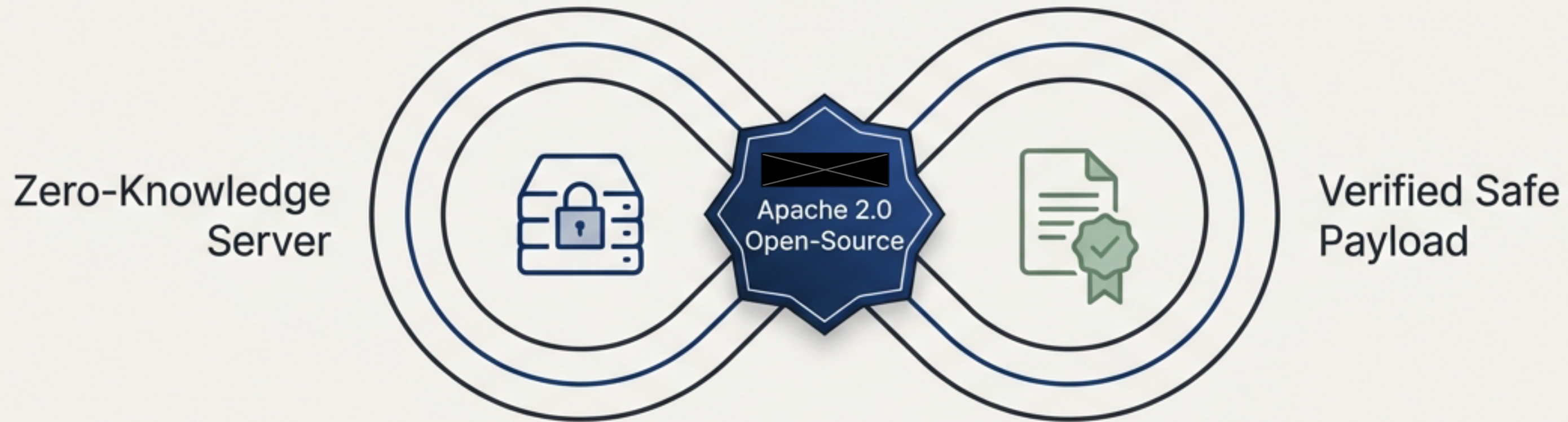
Content Trust & Native Signing. Partners adopt native cryptographic signing of their own scan results.





The Future of Verifiable File Transfer

A safe file, delivered safely.



- SG/Send solves the zero-knowledge paradox without compromising the network boundary.
- The receiver mandates the standard. The sender pays the micro-cost. The partner scales effortlessly.
- Built on an Apache 2.0 open-source foundation.

ENGAGE WITH THE V0.10.30 ARCHITECTURE AT SGRAPH.AI.